

Trainings ▾

General Information

Fuels used in Cummins® engines should conform to the specifications in Section 1 Diesel Fuel subsection (00-375-001) Diesel Fuel – Section Overview Products and Applications Approved for use of Military Fuels. **Only** fuels listed in the “Approved Fuel Types” column of the Products and Applications Approved for use of Military Fuels Table can be used in the intended application noted in Products and Applications Approved for use of Military Fuels. Fuels listed in Products and Applications Approved for use of Military Fuels Table **must** meet the appropriate fuel specification to be used successfully in Cummins® engines. Fuel property information listed in tables in this document are for reference **only**. Refer to the appropriate specification for any dispute or for further information about specific fuel properties.

Glossary of Fuel Types:

- JP-5 – Jet Propellant Grade 5 is a military grade kerosene fuel with a higher minimum flash point. This fuel is described by specification MIL-DTL-5624. Primary use of this fuel is in aircraft loaded on aircraft carriers.
- JP-8 – Jet Propellant Grade 8 is a military grade fuel that is similar to Jet A-1 with the addition of corrosion inhibitor and anti-icing additives. This fuel is described by specification MIL-T-83133. The primary uses for this fuel are military aircraft.
- F-75 – NATO designation for a type of naval distillate fuel. F-75 has similar physical and chemical properties versus F-76 but has a lower cloud point. Intended uses are for gas turbine power plants, high and medium speed compression ignition engines, and boilers.
- F-76 – NATO designation for a type of naval distillate fuel. F-76 has similar physical and chemical properties versus F-75 but has a higher cloud point. Intended uses are for gas turbine power plants, high and medium speed compression ignition engines, and boilers.
- F-24 – NATO designation for a type of aviation fuel which has similar physical and chemical properties to ASTM D1655 grade Jet A fuel. The Fuel System Icing Inhibitor (FSII) is different between Jet A and F-24.
- Jet-A – Civilian Aviation Fuel with similar fuel properties to Jet A-1, **only** differing by freeze point which is higher than Jet A-1.
- Jet A-1 – Civilian Aviation Fuel with similar fuel properties to Jet A, **only** differing by freeze point which is lower than Jet A.

Impacts to Engine Operation

Military fuels specifications vary across the world. Users should confirm that fuel intended to be used in engines conforms to Cummins Inc. requirements stated in this publication. Fuel property differences can affect the performance and durability of Cummins® engines when using JP-5,

JP-8, F-75, F-76, and F-24 fuels. Cetane number of JP-5 and JP-8 can be lower as compared to typical #2 ULSD (Ultra Low Sulfur Diesel). Cetane Numbers below 42 can cause issues such as difficulty starting at cold temperatures, excessive white smoke, and poor idling. Also of note, the lower energy content of JP-5 and JP-8 fuels could result in a 2 percent power reduction and slightly higher fuel consumption as compared to #2 ULSD (Ultra Low Sulfur Diesel). Lubricity enhancing filters or fuel additives are recommended when using these fuels due to poor lubricity. See application notes in Table 1 below. Use of fuels with poor lubricity will cause excessive wear in fuel system components leading to engine damage. Lubricity enhancing filters or multifunctional fuel additives such as Power Service Diesel Kleen +Cetane Boost can improve lubricity to Cummins Inc. minimum requirements per Table 1 in procedure 375-001 Diesel Fuel Overview. See Section 1. It is the customers responsibility to consult with fuel additive manufacturer to determine correct dosage for fuel treatment. Another risk of using aviation fuels such as JP-8 and F-24 is cavitation in the fuel system based on low initial boiling points. Injector durability could be affected by low boiling point fuels. JP-8 and F-24 fuel **must** conform to the requirements of Table Products and Applications Approved for use of Military Fuels . High sulfur levels of these fuels will cause damage to emissions control systems (if equipped). Based on environmental conditions and safety factors such as flash point of fuel, Cummins® recommends that fuel coolers should be fitted on engines using JP-5, JP-8, F-75, F-76, or F-24 fuels. Reference engine data sheets for flow rate and temperature information when selecting a fuel cooler.

Warranty and the Use of Military and Aviation Fuels in Cummins® Engines

Cummins Inc. Engine Warranty covers malfunctions that are a result of defects in material or factory workmanship. Engine damage, service issues, and/or performance issues determined by Cummins Inc. to be caused by using fuels **not** conforming to appropriate specifications are **not** considered to be defects in material or workmanship and are **not** covered under Cummins Inc. engine warranty. The use of military fuels in QSK engines **not** included in the table below, Products and Applications Approved for use of Military Fuels is contingent on fuels meeting the requirements in Table 1 in procedure 375-001 Diesel Fuel Overview. See Section 1.

Products and Applications Approved for use of Military Fuels

Product Series	Displacement (L)	Products	Approved Fuel Types	Specification Designation	Application	Application Notes
X	15	All Engines	JP-8 JP-5 F-76 F-75 F-24	MIL-DTL-83133 MIL-DTL-5624 MIL-DTL-16884 P29-4-3-11-2022 AFLP-3747	Marine and Defense only	Part number S20022 lubricity filter is required. See details: https://gce.cummins.com/acc_e_aebs/c_eb00331.pdf (https://gce.cummins.com/acc_e_aebs/c_eb00331.pdf)

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Products and Applications Approved for use of Military Fuels						
Product Series	Displacement (L)	Products	Approved Fuel Types	Specification Designation	Application	Application Notes
QSB	4.5, 5.9, 6.7, 7	All Engines	JP-8 JP-5 F-76 F-75 F-24	MIL-DTL-83133 MIL-DTL-5624 MIL-DTL-16884 P29-4-3-11-2022 AFLP-3747	Marine and Defense only	Part numberFS 20022 lubricity filter required for QSB5.9, and QSB7 engines. https://gce.cummins.com/ace_aws/cab00331.pdf

Products and Applications Approved for use of Military Fuels						
Product Series	Displacement (L)	Products	Approved Fuel Types	Specification Designation	Application	Application Notes
QSC	8.3	All Engines	JP-8 JP-5 F-76 F-75 F-24	MIL-DTL-83133 MIL-DTL-5624 MIL-DTL-16884 P29-4-3-11-2022 AFLP-3747	Marine and Defense only	(2) Part number FS20022 lubricity filters in parallel required. https://gce.cummins.com/ace_acebs/c eb00 331. pdf (https://gce.cummins.com/ace_acebs/c eb00 331. pdf)

Products and Applications Approved for use of Military Fuels						
Product Series	Displacement (L)	Products	Approved Fuel Types	Specification Designation	Application	Application Notes
QSL	9	All Engines	JP-8 JP-5 F-76 F-75 F-24	MIL-DTL-83133 MIL-DTL-5624 MIL-DTL-16884 P29-4-3-11-2022 AFLP-3747	Marine and Defense only	(2) Part number FS20022 lubricity filters in parallel required. https://gce.cu.mmins.com/ace_aebs/c eb00 331. pdf (https://gce.cu.mmins.com/ace_aebs/c eb00 331. pdf)
QSM	11	All Engines	JP-8 JP-5 F-76 F-75 F-24	MIL-DTL-83133 MIL-DTL-5624 MIL-DTL-16884 P29-4-3-11-2022 AFLP-3747	Marine and Defense only	

Products and Applications Approved for use of Military Fuels						
Product Series	Displacement (L)	Products	Approved Fuel Types	Specification Designation	Application	Application Notes
KTA	19, 38, 50	All Engines	JP-8 JP-5 F-76 F-75 F-24	MIL-DTL-83133 MIL-DTL-5624 MIL-DTL-16884 P29-4-3-11-2022 AFLP-3747	Marine and Defense only	Fuel must conform to all specification requirements and lubricity (HFRR) requirements listed in Table 1 of procedure 00-375-001, Section 1.
QSK	19, 38, 50, 60, 95	All Engines	JP-8 JP-5 F-76 F-75 F-24	MIL-DTL-83133 MIL-DTL-5624 MIL-DTL-16884 P29-4-3-11-2022 AFLP-3747	Marine and Defense only	Fuel must conform to all specification requirements and lubricity (HFRR) requirements listed in Table 1 of procedure 00-375-001, Section 1.

Products and Applications Approved for use of Military Fuels						
Product Series	Displacement (L)	Products	Approved Fuel Types	Specification Designation	Application	Application Notes
QSF	2.8, 3.8	QSF2.8: All Engines QSF3.8 Stage III A	JP-8	MIL-DTL-83133	Construction and Defense	Part Number FS20022 lubricity filter required. https://gce.cu.mmins.com/ace_aebs/c eb00 331. pdf (https://gce.cu.mmins.com/ace_aebs/c eb00 331. pdf)

Products and Applications Approved for use of Military Fuels						
Product Series	Displacement (L)	Products	Approved Fuel Types	Specification Designation	Application	Application Notes
ISB	6.7	Euro Stage III	JP-8	MIL-DTL-83133	Defense	Part number FS20022 lubricity filter required. https://gce.cummins.com/acc_e_aebs/c eb00331.pdf (https://gce.cummins.com/acc_e_aebs/c eb00331.pdf)

Products and Applications Approved for use of Military Fuels						
Product Series	Displacement (L)	Products	Approved Fuel Types	Specification Designation	Application	Application Notes
B	6.7	Stage V (Base Engine)	JP-8 JP-5 F-76 F-75 F-24	MIL-DTL-83133 MIL-DTL-5624 MIL-DTL-16884 P29-4-3-11-2022 AFLP-3747	Defense	Part number FS20022 lubricity filter required. https://gce.cummins.com/acc_e_aebs/c eb00 331. pdf (https://gce.cummins.com/acc_e_aebs/c eb00 331. pdf)

Products and Applications Approved for use of Military Fuels						
Product Series	Displacement (L)	Products	Approved Fuel Types	Specification Designation	Application	Application Notes
L	9	Euro Stage III	JP-8 F-24	MIL-DTL-83133 AFLP-3747	Defense	Part number FS20022 lubricity filter required. https://gce.cummins.com/ace_aebs/c eb00 331. pdf (https://gce.cummins.com/ace_aebs/c eb00 331. pdf)
F-75 / F-76 Fuels Properties						
Property		MIN		MAX		ASTM Test Method
Flash point, °C [°F]		60.0 [140]				D93, D3828, D7094
Kinematic viscosity at 40°C [104°F]		1.7		4.3		D445, D7042, D7945

F-75 / F-76 Fuels Properties			
Property	MIN	MAX	ASTM Test Method
Density at 15°C [59°F], kg/m ³ [lbs/ft ³] a. F-76 b. F-75	800.0 [49.938] 815.0 [50.879]	880.0 [54.937] 880.0 [54.937]	D1298 or D4052
Distillation			
90 percent Recovered, °C [°F]		357 [674.6]	D86 or D7345
Final boiling point, °C [°F]		385[725]	
Percent recovery, percent by volume	97		
Water and sediment, percent by volume		0.02	D1796 (modified) or D2709
Carbon residue on 10 percent bottoms, percent by mass		0.2	D189, D524, D4530
Ash, percent by mass		0.005	D482
Cloud Point, °C [°F] F-76 F-75 a. F-76 b. F-75		-1 [30.2] -12 [10.4]	D2500, D5771, D5772, D5773
Pour point, °C [°F] F-76 F-75 a. F-76 b. F-75		-6 [21.2] -18 [-0.4]	D97, or D5949
Cetane Number	40		D613, D6890, D7039
Sulphur, mg/kg		15	D2622, D5453, D7039
Strong acid number, mg KOH/g		Nil	D974
Acid number, mg KOH/g		0.30	D664 or D974
Storage stability, total insolubles, mg/100ml		1.5	D2274
Color, ASTM		3.0	D1500 or D6045

F-75 / F-76 Fuels Properties			
Property	MIN	MAX	ASTM Test Method
Copper strip corrosion, 3h at a minimum test temperature of 50°C [122°F]		No. 1	D130
Electrical conductivity, at point, time and temperature of delivery to purchaser, pS/m	25		D2624
Lubricity, avg. wear scar width, micrometers		460	D6074
Water Separability, separation time, min		10	D6217 or D7321
Particulate contamination, mg/L		10	D1401
Only for F-76 Content of synthetic hydrocarbon components, percent by volume, either: Synthesized paraffinic diesel and the remainder conventional hydrocarbons, or Synthesized iso-paraffins (farnesane) and the remainder conventional hydrocarbons		50 20	
Total aromatic hydrocarbons, percent by mass	8.1		D5186, D6591, D8368, or IP391
Trace Metals, mg/kg: Calcium Lead Sodium plus Potassium Vanadium		1.0 0.5 1.0 0.5	D7111, D3605

JP-8 Fuel Properties			
Property	MIN	MAX	ASTM Test Method
Color, Saybolt	Report		D156, or D6045
Total acid number, mg KOH/g		0.015	D3242
Aromatics, volume percent		25.0	D1319
Sulfur, total, mass percent		0.30	D129, D1266, D2622, D3129, D4294, or D5453
Sulfur, mercaptan, mass percent		0.002	D3227
Distillation temperature, °C [°F] Initial boiling point 10 percent recovered 20 percent recovered 50 percent recovered 90 percent recovered Final boiling point Residue, volume percent Loss, volume percent	Report Report Report Report	205 [401] 300 [572] 1.5 1.5	D86, D2887, D7345
Flash point, °C [°F]	38 [100.4]		D56, D93, D3828
Density, kg/L [lb/ U.S. gal] at 15°C [59°F] or Gravity, API at °C [60 °F]	0.775 [6.476] 37.0	0.840 [7.01] 51.0	D1298, D4052, D7777
Freezing point, °C [°F]		-47 [116.6]	D2386, D5972, D7153, or D7154
Viscosity, at -20 °C mm ² /s [°F]		8.0	D445, D7042
Net heat of combustion, MJ/kg	42.8		D3338, D4529, D4809
Hydrogen content, mass percent	13.4		D3343, D3701, D5291, or D7171

JP-8 Fuel Properties			
Property	MIN	MAX	ASTM Test Method
Smoke point, mm Or Smoke point, mm and Naphthalenes, volume percent	25.0 19.0	 3.0	D1322 D1322 D1840
Calculated Cetane Index	Report		D976 or D4737
Copper strip corrosion, 2 hr at 100 °C [°F]		No. 1	D130
Thermal stability (2.5 hr at 260 °C) \ [°F] Change in pressure drop, mm Hg Tube rating: One of the following requirements shall be met: Annex A1 VTRAnnex A3 ETR or Annex As ITR, average deposit thickness, nm, over area of 2.5mm ²		25 <3 85	D3241
Fuel system icing inhibitor, volume percent	0.07	0.10	D5006
Fuel electrical conductivity, pS/m	See MIL-DTL-83133J note 12		D2624
Existent gum, mg/100mL		7	D381 or IP540
Water reaction interface rating		1b	D1094
Microseparometer Rating	See MIL-DTL-83133J note 14		D3948 or D7224
Lubricity, average wear scar width, micrometers		>520	D6079
*Cummins Inc. requires 520 micrometers. If the fuel is >520, a lubricity filter or additive as specified in table 1 is required for use. This table requirement is not in MIL-DTL-83133.			

JP-5 Fuel Properties			
Property	Min	Max	ASTM Test Method
Color, Saybolt	Report		D156 or D6045
Total Acid number, mg KOH/g		0.015	D3242
Aromatics, volume percent		25.0 26.5	D1319 D6379
Sulfur, Mercaptan, mass percent		0.002	D3227
Sulfur, total, mass percent		0.20	D129, D1266, D2622, D3120, D4294, or D5453
Distillation temperature, °C [°F] Initial boiling point 10 percent recovered, temperature 20 percent recovered, temperature 50 percent recovered, temperature 90 percent recovered, temperature Final boiling point, temperature Residue, volume percent, (for D86) Loss, volume percent (for D86)	Report Report Report Report 1.5 1.5	205 [401] 300 [572]	D86, D7345, D2887
Flash point, °C [°F]	60.0 [140]		D56, D93, D3828
Density, at 15°C [59°F] kg/L (API gravity)	0.788 (48.0)	0.845 (36.0)	D1298, D4052, D7777
Freezing Point, °C [°F]		-46 [-50.4]	D2386, D5972, D7153, D7154
Viscosity, at -20 °C, mm ² /s [-4°F]		7.0	D445
Net Heat of combustion, MJ/kg	42.6		D3338/D3338M, D4529, or D4809
Calculated Cetane Index	Report		D976, D4737

JP-5 Fuel Properties			
Property	Min	Max	ASTM Test Method
Hydrogen content, mass percent	13.4		D3701, D5291, D7171
Smoke point, mm or Smoke point mm and Naphthalene, volume percent	25.0 18.0	3.0	D1322 D1322 D1840
Copper strip corrosion, 2hr at 100°C [212°F]		No.1	D130
Thermal stability (2.5 hr at 260°C [500°F]) Change in pressure drop, mm Hg Tube rating: One of the following requirements shall be met: Annex A1 VTR Annex A3 ETR or Annex As ITR, average deposit thickness, nm, over area of 2.5mm ²		25 <3 85	D3241
Existent gum, mg/100mL		7	D381
Micro-Separometer rating	See MIL-DTL-5624W		D3948
Fuel System icing inhibitor, volume percent	0.08	0.11	D5006
*Lubricity, average wear scar width, micrometers		520	D6079
*Cummins Inc. requires 520 micrometers. If the fuel is >520 micrometers, a lubricity filter or additive as specified in table 1 is required for use. This table requirement is not in MIL-DTL-83133.			

